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Oefeningenreeks 4A nr. 43.1

$$\int \frac{dx}{x(x-1)^2}$$

We splitsen de integraal op in partiële Breuken:

$$\frac{1}{x(x-1)^2} \Rightarrow \frac{A}{x} + \frac{B}{(x-1)^2} + \frac{C}{(x-1)}$$

$$A = \frac{1}{(x-1)^2}\Bigg|_{x=0} = 1$$

$$B = \frac{1}{x}\Bigg|_{x=1} = 1$$

$$\left.\begin{aligned} C &= \left(\frac{1}{x(x-2)^2} - \frac{B}{(x-1)^2}\right)\cdot(x-1) \\ &= \left(\frac{1-x}{x(x-1)^2}\right)\cdot(x-1) \\ &= \left(\frac{-(x-1)^2}{x(x-1)^2}\right) = -\frac{1}{x} \end{aligned}\right|_{x=1} = -1$$

Hier uit weten we nu dat $A = 1$, $B = 1$, $C = -1$

en we vullen het in de opsplitsing:

$$\Rightarrow \int \frac{A}{x}dx + \int \frac{B}{(x-1)^2}dx + \int \frac{C}{x-1}dx$$

$$\begin{aligned} &= \int \frac{1}{x}dx + \int \frac{1}{(x-1)^2}dx - \int \frac{1}{x-1}dx \\ &= \ln|x| + \int \frac{1}{(x-1)^2}dy - \int \frac{1}{x-1}dx \\ &= \ln|x| + \int y^{-2}dy - \int \frac{1}{y}dy \\ &= \ln|x| - \frac{1}{y} - \ln|y| + c \\ &= \ln|x| - \frac{1}{x-1} - \ln|x-1| + c \end{aligned}$$

$$y = x - 1 \Rightarrow x = y + 1$$
$$dy = dx$$

Je resultaat is dan:

$$\ln|x| - \frac{1}{x-1} - \ln|x-1| + c$$